The Alien Pebble Heuristic: A Framework for Detecting Non-Human Technosignatures in Low-Signal Astronomical Data

Executive Summary

This report presents a comprehensive framework for a novel heuristic in the Search for Extraterrestrial Intelligence (SETI), termed the "Alien Pebble Heuristic." The central thesis of this heuristic is that the most probable and widespread evidence of non-human technology will not manifest as deliberate, high-power communication signals but rather as subtle, unintentional artifacts or anomalies within voluminous, low-signal astronomical datasets. These "pebbles out of place" are defined not by their intrinsic energy but by their contextual improbability against a well-understood natural astrophysical background. The purpose of this document is to operationalize this conceptual heuristic into a robust, scientifically-grounded, and actionable research program.

The analysis begins by situating the heuristic within the historical context of SETI, arguing that the persistent lack of confirmed signals—the "Great Silence"—necessitates a paradigm shift away from signal-centric searches toward a broader, more inclusive search for all forms of "technosignatures." This report posits that the most effective methodology for this new paradigm is one rooted in unsupervised anomaly detection, a technique designed to identify "unknown unknowns" without prior training data. This approach structurally mirrors the evolution of advanced cybersecurity, which has moved from signature-based threat detection to more sophisticated anomaly-based systems capable of identifying novel threats.

A detailed taxonomy of potential "pebble" technosignatures is developed, drawing conceptual parallels with out-of-place artifacts in archaeology and biomarkers in diagnostic medicine. These analogies underscore a critical requirement: the ability to detect a contextual anomaly is contingent upon an exquisitely detailed, predictive model of the "natural baseline." Potential targets range from trace industrial pollutants in exoplanetary atmospheres and subtle thermal excesses to the anomalous kinetic behavior of interstellar objects and degraded

"technofossils" of extinct civilizations.

The report then outlines a practical computational framework for implementing the heuristic. This framework is architected as a multi-stage algorithmic funnel, designed to systematically process vast astronomical data streams from surveys like the Vera C. Rubin Observatory and the Square Kilometre Array. The funnel employs a sequence of unsupervised machine learning algorithms—from broad screening methods like Local Outlier Factor to deep learning models like Autoencoders and temporal analyzers like Long Short-Term Memory networks—to progressively isolate and build confidence in candidate anomalies.

A significant portion of the analysis is dedicated to addressing the profound statistical challenges inherent in such a search. The report provides a rigorous examination of the "look-elsewhere effect," a statistical pitfall where searching large parameter spaces guarantees the discovery of spurious signals. To counter this and manage the inevitable high rate of false positives, a sophisticated vetting process is proposed, culminating in a human-in-the-loop active learning system. This collaborative human-machine approach is designed not only to refine the search for technosignatures but also to create a positive feedback loop that systematically discovers and classifies new and rare natural astrophysical phenomena, providing a dual scientific return on investment.

Ultimately, this report concludes that the Alien Pebble Heuristic represents more than a new search strategy; it is a fundamental re-framing of SETI as a form of cosmic archaeology. The search is no longer for a voice but for a trace; not for a message, but for an artifact. The detection of such a "pebble" would have profound scientific and philosophical implications, challenging humanity's understanding of its place in the universe and the ultimate fate of technological civilizations. The recommendations provided herein outline a clear, data-driven path toward making this ambitious search a reality.

Section 1: Introduction: A Paradigm Shift in the Search for Extraterrestrial Intelligence

1.1 From Beacons to Breadcrumbs: The Limitations of Signal-Centric SETI

The modern scientific Search for Extraterrestrial Intelligence (SETI) began with a clear and compelling premise. In 1960, astronomer Frank Drake initiated Project Ozma, using a radio

telescope to listen for signals from nearby Sun-like stars.[1] The experiment was predicated on a set of logical, albeit anthropocentric, assumptions: that an extraterrestrial intelligence (ETI) would be actively attempting to communicate across interstellar distances, that they would choose an efficient and durable medium like radio waves, and that they would encode their message in a format that was clearly artificial and distinguishable from the cacophony of natural cosmic noise, such as a narrow-band signal.[1] This signal-centric paradigm, the search for a deliberate beacon, has defined the mainstream of SETI research for over half a century.

Despite decades of increasingly sophisticated searches and the deployment of advanced digital receivers capable of monitoring millions of channels simultaneously, the endeavor has been met with a profound and persistent silence.[1] While tantalizing candidate signals have occasionally emerged, such as the famous "Wow!" signal detected in 1977, none have been confirmed upon subsequent observation, highlighting the immense difficulty of verifying transient, non-repeating events.[1] This null result, often referred to as the "Great Silence," is the observational cornerstone of the Fermi Paradox: the stark contradiction between the high probability estimates for the existence of ETI and the complete lack of evidence for their presence.[5] The failure of the signal-centric approach to yield a confirmed detection has served as a primary driver for a critical re-evaluation of its foundational assumptions and a search for alternative strategies.

1.2 The Emerging Science of Technosignatures: A New Search Space

In response to the limitations of the traditional paradigm, the scope of SETI has expanded significantly to encompass the search for "technosignatures." A technosignature is formally defined as any measurable property, effect, or artifact that provides scientific evidence of past or present technology.[8] This definition represents a crucial conceptual leap, moving beyond the narrow search for deliberate communication to a broader hunt for any detectable manifestation of technology. It marks a shift from seeking a "message in a bottle" to performing a comprehensive forensic analysis of the cosmos.

The modern SETI landscape now includes searches for a diverse array of potential technological evidence. These range from the grandiose and speculative, such as astroengineering megastructures like Dyson spheres designed to harvest a star's total energy output, to far more subtle indicators.[4] Researchers are now actively considering how to detect the atmospheric pollution generated by an alien industry (e.g., chlorofluorocarbons or nitrogen dioxide), the waste heat emanating from planetary-scale industrial processes, or the faint, persistent glow of city lights on the night side of an exoplanet.[4] The common thread among these new targets is a focus on detecting the

unintentional byproducts of a technological civilization. The search is no longer solely for a deliberate broadcast but for the cosmic "spillover" or "leakage" of a civilization simply existing and manipulating its environment.[11]

This evolution in SETI methodology mirrors a similar and instructive paradigm shift in the field of cybersecurity. Early antivirus software operated on a signature-based model, scanning for the known digital "fingerprints" of existing viruses.[16] This approach was effective against known threats but was powerless against new, "zero-day" attacks for which no signature yet existed. To combat this, the field developed anomaly-based intrusion detection systems (IDS). These systems do not look for specific threats; instead, they build a statistical baseline of "normal" network behavior and flag any significant deviation from that baseline as a potential intrusion.[16] Traditional SETI, with its search for specific, pre-defined signal types like narrow-band radio waves, is analogous to signature-based detection. The move to a broader search for technosignatures is a philosophical and methodological shift toward an anomaly-based framework. It is an acknowledgment that we do not know what an alien signal or artifact will look like, so the most effective strategy is to search for

anything that deviates from the natural astrophysical background.

1.3 The Alien Pebble Heuristic: Defining a Search for Subtle, Contextual Anomalies

This report formally proposes and develops a specific strategy within this new anomaly-based paradigm: the "Alien Pebble Heuristic." The heuristic is defined as a targeted search methodology for identifying non-human technological patterns by focusing on tiny, statistically subtle, and otherwise irrelevant anomalies—"pebbles out of place"—within voluminous, low-signal astronomical data.

The core premise of the heuristic is that the most common, persistent, and widespread technosignatures will not be powerful, energy-intensive beacons but rather subtle, low-level byproducts of technological activity. These "pebbles" are anomalous not because of their high energy or statistical significance in isolation, but because of their *contextual improbability*. They are the cosmic equivalent of finding a modern-day wristwatch in a sealed Egyptian tomb; the object itself is unremarkable, but its presence in that specific context is a profound anomaly that demands explanation. The term "heuristic" is deliberately chosen, drawing from its use in computer science and cybersecurity, where it denotes a practical, experience-based "rule of thumb" used for problem-solving and threat detection when exact, algorithmic solutions are computationally infeasible or unknown.[18] The Alien Pebble Heuristic is thus a practical guideline for prioritizing search efforts in the astronomically large

parameter space of potential technosignatures.

The Fermi Paradox itself provides a powerful justification for this focus on subtlety. The fact that we have not detected obvious, galaxy-spanning engineering projects (e.g., Kardashev Type II or III civilizations) strongly suggests that such overt manifestations of technology are either extremely rare, transient, or their signatures are not what we expect.[4] This null result from searches for "macro-signatures" acts as a powerful constraint, effectively lowering the probability of encountering high-power, easily detectable technosignatures. This logically compels a shift in focus toward the lower-power, subtler end of the detectability spectrum. The paradox is not merely a motivation to search differently; it is an empirical observation that guides the search toward the small, the quiet, and the contextually strange. The "pebbles" are precisely what one might expect to find in a universe that appears, at first glance, to be silent.

Section 2: The Nature of the Pebble: A Taxonomy of Low-Observable Technosignatures

2.1 Unintentional Leakage and Incidental Artifacts: The Most Probable Alien Traces

A foundational assumption of the Alien Pebble Heuristic is that any sufficiently advanced and long-lived technological civilization will inevitably and unintentionally alter its environment in ways that are detectable over interstellar distances.[14] These alterations are not messages; they are byproducts, the incidental exhaust of a functioning technosphere. The most accessible proxy for modeling such signatures is our own planet. The current geological epoch, the Anthropocene, is defined by humanity's planetary-scale impact, making Earth itself a source of potent technosignatures.[21]

One of the most-discussed examples is electromagnetic leakage. Civilian and military radar systems, for instance, emit powerful, structured radio signals that are not intended for interstellar communication but leak into space nonetheless. Recent studies suggest that the combined signal from Earth's airport and military radar systems could be detectable by a civilization with technology comparable to our own from as far as 200 light-years away.[12] These signals would appear clearly artificial due to their narrow bandwidth and structured, repetitive nature, serving as an unambiguous, albeit unintentional, beacon of our

technological presence.

Beyond radio leakage, a civilization's industrial and domestic activities could produce a host of other detectable artifacts. The chemical composition of an exoplanet's atmosphere, observable via transmission spectroscopy, could hold compelling evidence. The presence of complex, artificial molecules like chlorofluorocarbons (CFCs) or significant, localized concentrations of simpler industrial pollutants like nitrogen dioxide (NO2) would be difficult to explain through natural geological or biological processes alone.[8] Similarly, the energy usage of a global civilization could manifest as a persistent, low-level thermal excess, akin to a planetary-scale "urban heat island" effect, detectable in the infrared spectrum.[4] On the night side of a tidally locked or slowly rotating exoplanet, the collective glow of artificial lighting from cities and infrastructure could create a unique and distinguishable spectral signature, particularly if dominated by specific lighting technologies like sodium-vapor lamps.[4]

2.2 Analog from Archaeology: Out-of-Place Artifacts (OOPArts) as a Model for Contextual Discovery

The concept of an "out-of-place artifact" (OOPArt) from archaeology provides a powerful intellectual framework for understanding the discovery process envisioned by the Alien Pebble Heuristic.[23] An OOPArt is an object of historical or archaeological interest found in a context that challenges the conventional historical or technological timeline. The anomaly is not necessarily the object itself, but its

presence in that specific context.[23]

Consider the Antikythera mechanism, a highly complex mechanical computer, an orrery, recovered from a shipwreck dating to the 2nd century BCE. The device exhibits a level of miniaturization and mechanical sophistication that was not thought to exist until the 14th century CE.[24] The mechanism is anomalous because it is technologically out of place. Another example is the Maine penny, an 11th-century Norwegian coin found in a Native American archaeological site in Maine, thousands of miles from any known Viking settlement in North America. The coin itself is mundane, but its discovery in that location strongly suggests a pre-Columbian contact or trade network that is otherwise poorly documented.[24]

Extending this analogy to the cosmos, the Alien Pebble Heuristic is fundamentally a search for cosmic OOPArts. The "pebble" could be an interstellar object, like the enigmatic 'Oumuamua, that exhibits slight non-gravitational acceleration not easily explained by natural outgassing.[25] It could be an asteroid with an unnatural spectrum, suggesting an artificial coating or composition. It might be a brief, structured, but non-repeating radio transient from a region of

the sky with no known energetic astrophysical sources. In each case, the object or signal is a "pebble out of place," an anomaly defined by its deviation from the expected context. This reframes SETI as a form of "archaeology of extraterrestrial intelligence," a discipline focused on reconstructing past or present technological activity from fragmentary and contextually anomalous evidence.[2] This archaeological perspective naturally leads to the concept of "technofossils." While unintentional leakage points to active technology, a technofossil is the degraded remnant of technology that may no longer be functional. Examples could include a belt of defunct satellites (a "Clarke Exobelt") orbiting an exoplanet, the lingering atmospheric signature of a long-past industrial era, or even the physical debris from interstellar craft.[8] The search for technofossils vastly expands the temporal window for detection, allowing for the discovery of civilizations that may have existed millions or even billions of years ago.

2.3 Analog from Medicine: Biomarkers as a Model for Early, Low-Level System Indicators

A second powerful analogy for the heuristic comes from the field of modern diagnostic medicine, specifically the use of biomarkers. A biomarker is a measurable indicator of a biological state or condition, often a molecule or protein found in bodily fluids.[28] Their primary value lies in the early detection of disease, often long before the emergence of overt clinical symptoms.[28] For example, elevated levels of the protein CA-125 can be an early indicator of ovarian cancer, while the presence of troponin in the blood is a crucial marker for a heart attack.[28]

This medical model frames the search for "pebbles" as a form of cosmic diagnostic screening. A complex system—in this case, a technological civilization—can be detected via its subtle, low-level chemical or physical traces. The presence of CFCs in an exoplanet's atmosphere is not just pollution; it is a "diagnostic biomarker" for the "condition" of active industrial technology.[22] A localized thermal anomaly is a biomarker for significant energy consumption. This approach emphasizes detection at the earliest and most subtle stages, where the signal may be weak but is nonetheless unambiguous if the "baseline" of a healthy, non-technological planet is well understood.

This analogy highlights a critical prerequisite for the success of the Alien Pebble Heuristic. Just as a doctor cannot interpret a blood test without a detailed understanding of normal human physiology, an astronomer cannot identify a technosignature without an exquisitely detailed, predictive model of the "natural baseline." The search for a pebble out of place is impossible without a perfect map of where all the natural pebbles are supposed to be. Therefore, the greatest challenge for the heuristic is not merely the development of more sensitive anomaly detection algorithms, but the parallel development of more sophisticated

and accurate simulations of the null hypothesis—that is, the physics, chemistry, and geology of a universe devoid of technology. The search for alien intelligence thus becomes an inseparable driver for fundamental research in astrophysics and planetary science.

To operationalize this search, the following table provides a structured taxonomy of potential "pebble" technosignatures, categorizing them by physical domain and outlining the observational requirements and key challenges for each.

Table 1: Taxonomy of Potential "Pebble" Technosignatures

Signature Category	Specific Example ("Pebble")	Primary Data Source	Key Detection Challenge
Atmospheric	Presence of artificial pollutants (e.g., CFCs, NO2) or unnatural chemical disequilibrium (e.g., O2 and CH4 co-existing without a clear biological source).[8]	High-resolution transmission and emission spectra from space telescopes (e.g., JWST, Habitable Worlds Observatory).	Distinguishing from unknown natural geochemical or photochemical pathways; achieving sufficient SNR to detect trace gases.[22]
Kinetic	Non-gravitational acceleration or anomalous orbital characteristics of interstellar objects, asteroids, or exoplanets.[25]	High-cadence, long-baseline photometric surveys (e.g., Vera C. Rubin Observatory/LSST) to generate precise light curves and astrometry.	Conclusively ruling out natural causes such as outgassing, radiation pressure, or complex, unmodeled N-body interactions.[25]
Thermal	Localized, stable, low-level thermal excess on a planetary surface, inconsistent with natural geothermal or insolation patterns (i.e., an	High-resolution infrared photometry and mapping from space telescopes (e.g., Spitzer, JWST).	Differentiating from natural sources like volcanism or unusual surface albedo properties; achieving sufficient spatial resolution.[4]

	"urban heat island").[4]		
Electromagnetic	Unintentional, narrow-band, structured radio or laser signal leakage (e.g., from radar, propulsion, or power beaming).[12]	Deep archival data from radio telescope surveys (e.g., SKA, FAST, Green Bank Telescope) and optical surveys (e.g., Laser SETI).	Excising pervasive radio frequency interference (RFI) from terrestrial sources; verifying the extraterrestrial origin of transient signals.[3]
Surface Artifact	Unnatural geometric shapes, patterns, or albedo variations on a planetary or lunar surface (e.g., large-scale engineering, mining operations).[8]	High-resolution direct imaging or reflected light analysis from planetary probes or future direct-imaging space telescopes.	Overcoming limited spatial resolution; avoiding the psychological trap of pareidolia (perceiving patterns in random data).[27]
Technofossil	A defunct belt of artificial satellites ("Clarke Exobelt") creating an anomalous transit light curve signature.[8]	Ultra-high-precisio n photometry from transit surveys (e.g., Kepler, TESS, PLATO).	The signature is extremely subtle and requires immense precision; distinguishing it from natural phenomena like planetary rings.[8]

Section 3: The Search Space: Characterizing the Astronomical Data Haystack

3.1 The Data Deluge: An Overview of Relevant Astronomical Surveys

The "voluminous data" component of the Alien Pebble Heuristic is not a future projection but a present-day reality. Modern astronomy is characterized by a data deluge, with large-scale surveys generating petabytes of information and creating unprecedentedly vast and complex datasets.[32] These surveys are the "haystacks" in which the heuristic proposes to search for anomalous "needles."

Key instruments and surveys providing the raw material for this search include:

- **Optical Sky Surveys:** The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) will image the entire visible southern sky every few nights for a decade, generating roughly 20 terabytes of data nightly. Its high-cadence, multi-band photometry will produce exquisitely detailed light curves for billions of objects, making it an ideal resource for detecting kinetic and transient "pebbles".[33]
- **Radio Telescopes:** The Square Kilometre Array (SKA), when fully operational, will be the world's largest radio telescope, with a sensitivity and survey speed orders of magnitude greater than current instruments. It will generate more data in a single day than the current total daily internet traffic. The SKA, along with existing facilities like China's FAST and the Green Bank Telescope, provides the deep radio archives necessary to search for electromagnetic leakage and other structured transients.[3]
- **Space-Based Observatories:** Space telescopes provide access to wavelengths blocked by Earth's atmosphere and deliver unparalleled precision. The James Webb Space Telescope (JWST) is currently providing high-resolution spectra of exoplanet atmospheres, the primary data source for detecting atmospheric technosignatures.[34] Archival data from missions like Kepler and the Transiting Exoplanet Survey Satellite (TESS) offer a wealth of high-precision photometric data for searching for transit anomalies like technofossils.[32]

This mapping of data sources to potential "pebbles" demonstrates that the necessary observational capabilities are, in large part, already in place or coming online. The primary challenge is no longer data acquisition but data analysis: developing the computational tools to effectively mine these immense and complex haystacks.

3.2 The Physics of Faintness: Understanding and Mitigating Low Signal-to-Noise Ratios (SNR)

The search for "pebbles" is fundamentally a search in a low-signal environment. The Signal-to-Noise Ratio (SNR or S/N) is a critical measure in science and engineering that compares the level of a desired signal to the level of background noise.[36] It is formally defined

as the ratio of signal power to noise power:

SNR=PnoisePsignal
where P is the average power of the signal and noise, respectively.[36] This is often expressed in decibels (

dB) using a logarithmic scale:

SNRdB=10log10(PnoisePsignal)
A high SNR indicates a clear, easily detectable signal, while a low SNR means the signal is obscured by noise and difficult to distinguish.[36] For many applications in imaging, the Rose criterion states that an SNR of at least 5 is required to distinguish features with high certainty.[36] The faintness of potential technosignatures from interstellar distances means they will almost certainly be in the low-SNR regime.

Therefore, a core technical component of the Alien Pebble Heuristic must be the application of techniques to enhance SNR. Several methods are available:

- **Filtering:** If the characteristics of the noise are known and differ from the signal (e.g., broadband noise vs. narrow-band signal), a filter can be used to reduce the noise. Advanced techniques like using a lock-in amplifier can extract a narrow bandwidth signal from noise that is a million times stronger.[36]
- **Averaging:** When a signal is constant or periodic and the noise is random, averaging multiple measurements can significantly enhance the SNR. The signal component adds linearly with the number of measurements (N), while the random noise adds in quadrature, meaning the noise level decreases as the square root of N. This is a powerful technique for teasing out faint, persistent signals from a noisy background.[36]
- **Frequency Domain Analysis:** Techniques like the Discrete Fourier Transform (DFT) can decompose a time-domain signal into its constituent frequencies. This is particularly effective for separating a structured, narrow-band artificial signal from broadband, stochastic astrophysical noise.[37]

Paradoxically, the condition of a low-SNR environment may actually favor the detection of artificial signals. Natural astrophysical radio emissions are typically broadband in nature.[3] In contrast, many potential artificial signals, even unintentional ones like radar leakage, are inherently narrow-band and structured.[3] In a noisy environment, a weak but spectrally concentrated (narrow-band) signal can be more easily isolated and its SNR enhanced through filtering and frequency analysis than a weak but spectrally diffuse (broadband) natural signal. The very problem of a low-SNR environment may thus act as a natural filter that preferentially highlights the kind of information-rich, structured signals that technology produces.

3.3 The Curse of Dimensionality in Astronomical Datasets

A formidable challenge facing any anomaly detection algorithm in astronomy is the "curse of dimensionality." This term describes the counter-intuitive phenomena that occur when analyzing data in high-dimensional spaces.[38] As the number of features or dimensions increases, the volume of the space increases so rapidly that the available data become sparse. In this high-dimensional space, every data point can appear to be an outlier, making it difficult for algorithms to distinguish meaningful anomalies from the background sparsity.[38]

Astronomical datasets are often inherently high-dimensional and multivariate. A single stellar spectrum from a modern instrument can contain thousands of frequency bins, each acting as a dimension. A time-series light curve from a survey like LSST will have measurements in multiple color bands taken at thousands of time steps, creating a very high-dimensional feature space for each object.[38] When traditional anomaly detection methods, which were often developed for low-dimensional data, are applied directly to such datasets, they can suffer from prohibitive computational costs and may produce invalid or misleading results.[38]

This "curse" has a profound implication for the strategy of the Alien Pebble Heuristic. It suggests that a search for anomalies within a single, monolithic, high-dimensional dataset is likely to be inefficient and plagued by false positives. The most robust way to confirm a "pebble" and overcome the statistical ambiguity of high-dimensional space is through the fusion of heterogeneous data sources. A candidate anomaly identified in one dataset (e.g., a peculiar transient in a TESS light curve) is, on its own, statistically weak. However, if that same object exhibits a corresponding, co-temporal anomaly in a completely different physical domain (e.g., a simultaneous radio burst detected by FAST or a unique spectral feature observed by JWST), the joint probability of this being a natural coincidence drops precipitously. The solution to the curse of dimensionality is therefore not just the application of more sophisticated algorithms, but the implementation of a multi-messenger and multi-modal data fusion strategy.[32] The Alien Pebble Heuristic must be architected to correlate anomalies across disparate observational platforms, seeking not just single points of strangeness, but coherent patterns of anomalous behavior across the electromagnetic spectrum and over time.

Section 4: The Heuristic Engine: A Computational Framework for Anomaly Detection

4.1 Unsupervised Learning as the Core Methodology for Finding "Unknown Unknowns"

The computational core of the Alien Pebble Heuristic must be built upon unsupervised machine learning. This is a categorical necessity dictated by the nature of the search. Supervised learning methods require large, pre-labeled training datasets to build a classification model.[39] For example, to train a supervised model to find alien signals, one would need a dataset containing numerous examples of both alien signals and non-alien signals. This is a logical impossibility, as we have no confirmed examples of the target class; we are searching for "unknown unknowns".[40]

Unsupervised learning, in contrast, operates without labeled data. Its primary goal in the context of anomaly detection is to learn the intrinsic structure and statistical properties of the available data—assumed to be overwhelmingly "normal"—and then identify instances that deviate significantly from this learned model.[40] This approach aligns perfectly with the "pebble out of place" philosophy. The algorithm first learns what it means for a celestial object or signal to be "in place" by modeling the vast corpus of normal astrophysical phenomena. It then flags any object that does not conform to this model as a potential "pebble," a candidate for further investigation. This makes unsupervised learning the only viable methodology for a truly agnostic search for novel technosignatures.

4.2 A Multi-Stage Algorithmic Funnel

A practical implementation of the heuristic requires more than a single algorithm. Given the scale of the data and the subtlety of the targets, a multi-stage algorithmic funnel is proposed. This architecture is designed to efficiently sift through massive datasets, progressively refining the pool of candidates and increasing the confidence in their anomalous nature. The funnel moves from computationally inexpensive, broad-stroke methods to more computationally intensive, specialized analyses.

Stage 1: Broad Screening (Density and Proximity-Based Methods)
The first stage aims to perform an initial, rapid triage of the entire dataset. The goal is to identify the most obvious outliers based on their position within the data's feature space.

- **Clustering Methods:** Algorithms like DBSCAN (Density-Based Spatial Clustering of Applications with Noise) are effective at grouping data points into clusters of similar objects. Any data point that does not belong to a cluster is naturally classified as noise or

an outlier, making it a candidate anomaly.[42]

- **Proximity-Based Methods:** Algorithms like k-Nearest Neighbors (KNN) and Local Outlier Factor (LOF) operate on the principle that normal data points exist in dense neighborhoods, while anomalies are isolated. LOF, for instance, calculates an anomaly score for each data point by comparing its local density to the local densities of its neighbors.[43] Streaming variants like MiLOF have been developed to apply this logic to continuous data streams, making them suitable for real-time analysis of survey data.[43] These methods are excellent for a first-pass filter.

Stage 2: Feature Extraction and High-Dimensional Analysis (Subspace and Reconstruction Methods)
Candidates that pass the initial screening are then subjected to more sophisticated algorithms designed to handle the curse of dimensionality and detect more subtle, structural anomalies.

- **Subspace Methods:** Isolation Forest (iForest) is a particularly efficient algorithm for this stage. Unlike methods that build a complex model of normal data, iForest works by randomly partitioning the feature space with decision trees. The logic is that anomalies are "few and different," and therefore they will be easier to isolate in fewer partitions than normal points.[44] Its efficiency makes it well-suited for large astronomical catalogs.
- **Reconstruction-Based Methods:** Deep learning models, particularly Autoencoders (AEs) and Variational Autoencoders (VAEs), are powerful tools for this stage.[40] An autoencoder is a type of neural network trained to learn a compressed, low-dimensional representation of its input data (encoding) and then reconstruct the original data from that representation (decoding). When trained exclusively on "normal" data (e.g., images of typical galaxies), the network becomes very good at reconstructing them. When presented with an anomalous input (e.g., an image of a merging galaxy or an object with an artificial structure), it will struggle to reconstruct it accurately. The magnitude of this "reconstruction error" serves as a robust anomaly score. This method is highly effective for complex data types like images and spectra.

Stage 3: Temporal and Sequential Analysis (Time-Series Methods)
For data that evolves over time, such as light curves from photometric surveys or dynamic spectra from radio observations, a final stage of temporal analysis is required.

- **Recurrent Neural Networks (RNNs):** Models like Long Short-Term Memory (LSTM) networks are specifically designed to learn patterns and dependencies in sequential data. An LSTM can be trained on a vast library of normal stellar light curves to learn the typical patterns of variability. It can then be used to predict the future behavior of a new light curve. A significant deviation between the model's prediction and the actual observation would be flagged as a temporal anomaly—a potential transient event, an unusual eclipse, or some other unexpected behavior.[44]
- **Forecasting Frameworks:** Statistical frameworks like Facebook's Prophet are also designed for time-series forecasting and anomaly detection. They can model complex seasonalities and trends, making them effective tools for identifying unexpected events in

long-term monitoring data.[44]

This funnel architecture provides an epistemological framework for managing uncertainty. It is designed not just to find anomalies, but to build confidence in their significance. An object that is flagged in Stage 1 is merely "different." If it also passes Stage 2, it is "structurally complex and different." If it then passes Stage 3, it is "structurally complex, different, and behaves unexpectedly over time." An anomaly that survives the entire funnel is no longer just a statistical outlier; it is a persistent, structurally complex, and dynamically unusual phenomenon, making it a high-priority candidate for a technosignature.

4.3 Analog from Cybersecurity: Anomaly-Based Intrusion Detection as a Technical Blueprint

The architecture of an anomaly-based Intrusion Detection System (IDS) in cybersecurity provides a direct and practical blueprint for the Alien Pebble Heuristic's operational framework.[16] The process is functionally identical:

1. **Establish Baseline:** The IDS first analyzes a large volume of normal network traffic to build a statistical model of what constitutes "normal" behavior. In the astronomical context, this corresponds to training the unsupervised models in the funnel on vast archives of existing astronomical data to learn the characteristics of natural phenomena.
2. **Monitor in Real-Time:** The IDS continuously monitors live network packets, comparing them against the established baseline. The astronomical equivalent is the continuous ingestion of new data streams from surveys like LSST or SKA.
3. **Flag Deviations:** Any activity that significantly deviates from the normal baseline is flagged as a potential intrusion and an alert is generated. For the heuristic, any data point that is flagged by the algorithmic funnel as a high-confidence anomaly becomes a candidate "pebble."

Furthermore, the distinction in cybersecurity between static and dynamic heuristic analysis offers a useful conceptual extension.[18] Static analysis involves examining the source code of a program without running it. This is analogous to analyzing a single astronomical data point in isolation (e.g., a single spectrum or image). Dynamic analysis involves executing the suspicious program in a safe, isolated "sandbox" environment to observe its behavior. This is analogous to taking a candidate "pebble" and using it as the initial condition for a series of physical simulations. If the object's future evolution under known physical laws diverges significantly from subsequent observations, it strengthens the case for its anomalous nature.

The following table provides a technical comparison of the key unsupervised algorithms proposed for the heuristic engine, outlining their strengths, weaknesses, and primary

applications in the search for "pebbles."

Table 2: Comparison of Unsupervised Anomaly Detection Algorithms for Astronomical Data

Algorithm/Class	Principle of Operation	Strengths for "Pebble" Detection	Weaknesses/Challenges	Primary Data Application
Isolation Forest (iForest)	Isolates anomalies through random partitioning in decision trees. Anomalies require fewer partitions to be isolated.[43]	Highly efficient on large datasets; scales well with high dimensionality; requires no distance calculations.[43]	Can be less effective at detecting anomalies in dense clusters (global anomalies); performance can depend on the number of trees.[43]	Large astronomical catalogs (e.g., Gaia, SDSS); feature-extracted data.
Autoencoder (AE/VAE)	A neural network learns a compressed representation of normal data and flags inputs with high reconstruction error as anomalous.[40]	Excellent at learning complex, non-linear patterns in high-dimensional data; effective dimensionality reduction; no assumptions about data distribution.[40]	Requires significant data and computational resources for training; can be a "black box," making results difficult to interpret; sensitive to hyperparameters.[43]	Image data (e.g., galaxy surveys); spectral data; multi-band photometry.
Local Outlier Factor (LOF)	Measures the local density deviation of a data point with respect to its neighbors.	Does not require any assumptions about the underlying data	High computational complexity (O(n2)), making it slow on large, static	Clustered survey data; initial screening of smaller, targeted

	Outliers have a significantly lower density than their neighbors.[43]	distribution; effective at finding local anomalies that other methods might miss.[43]	datasets; performance degrades in very high dimensions.[43]	datasets.
LSTM (Long Short-Term Memory)	A type of recurrent neural network that models sequential dependencies and can predict future values in a time series. Anomalies are deviations from predictions.[44]	Specifically designed for temporal data; can capture complex, long-range dependencies; robust to noise in time-series data.	Computationally intensive to train; can be difficult to tune; interpretation of the model's internal state can be challenging.	Photometric light curves (e.g., TESS, Kepler); dynamic spectra (radio transients).

A notable aspect of the deep learning models in this framework is their "black box" nature, which is often cited as a weakness due to a lack of interpretability.[43] However, in the context of a search for "unknown unknowns," this can be transformed into a strength. When an autoencoder flags a galaxy image as anomalous, the initial reason may not be clear. This forces a new mode of scientific inquiry where the algorithm does not test a pre-existing hypothesis but instead

generates a new one. The task for the human scientist then becomes one of interpretation: "What specific feature of this object did my model fail to learn from the millions of 'normal' examples, and is that feature a known but rare natural phenomenon, or is it something else entirely?" In this way, the algorithm becomes a true discovery engine, pointing researchers toward phenomena they would not have conceived to look for, which is the ultimate goal of the heuristic.

Section 5: The Perils of Discovery: Ensuring Statistical Rigor

5.1 The Look-Elsewhere Effect: Avoiding the Trap of Spurious Significance

The single greatest statistical peril facing the Alien Pebble Heuristic is the "look-elsewhere effect".[49] This phenomenon arises when one searches a large parameter space for a signal or anomaly. The sheer size of the search space makes it statistically certain that random fluctuations of noise will, by pure chance, conspire to create seemingly significant signals.[50] A statistical test might yield a p-value—the probability of observing a result at least as extreme if no real effect is present—that is very low for a specific location in the parameter space. However, when one has "looked elsewhere" at millions or billions of other locations, finding such a low p-value somewhere is not surprising, but expected.[50]

This effect is a notorious source of false discoveries in science. A classic example is a 1992 Swedish study that examined over 800 different ailments in populations living near high-voltage power lines. It found a statistically significant four-fold increase in childhood leukemia, sparking public alarm. However, the study failed to correct for the look-elsewhere effect; when testing 800 independent hypotheses, one is highly likely to find a "significant" result with a p-value of 1/800 by chance alone. Subsequent, more rigorous studies found no such link.[50] The application to SETI is direct and sobering. When scanning billions of radio channels or analyzing light curves from millions of stars, one is

guaranteed to find anomalies that look like the "Wow!" signal.[1] The fundamental challenge is not finding these candidates, but rigorously demonstrating that they are not simply statistical artifacts generated by the vastness of the search.

To combat this, statistical corrections must be applied. The simplest is the Bonferroni correction, which adjusts the significance threshold (α) by dividing it by the number of independent tests performed (n).[50] More sophisticated methods have been developed that relate the statistical significance to prior and posterior volume ratios in a Bayesian context, providing a more robust way to quantify the global p-value without resorting to computationally expensive simulations.[51] Any candidate "pebble" must have its statistical significance evaluated not in isolation (the "local" p-value) but in the context of the entire search that was conducted (the "global" p-value).

5.2 Managing the False Positive Epidemic: A Framework for Candidate Vetting

The practical output of any anomaly detection system is a list of candidates, the vast majority of which will be false positives (also known as Type I errors).[16] A false positive occurs when the system incorrectly classifies a normal data point as anomalous.[53] In the context of the Alien Pebble Heuristic, these are natural objects or instrumental effects that are sufficiently unusual to be flagged by the algorithms.

Managing this inevitable flood of false positives is a critical operational challenge. Each false positive represents a significant cost in terms of limited and expensive resources, such as follow-up observation time on world-class telescopes and the analytical time of human experts.[31] A system that produces too many false alarms (a high False Positive Rate, or FPR) will quickly become unusable, overwhelming the resources available for vetting.[16]

A multi-pronged strategy is required to mitigate this problem:

- **Threshold Adjustment:** Most anomaly detection algorithms produce a continuous anomaly score rather than a binary label. The classification threshold can be adjusted to make the system more or less sensitive. A higher threshold will reduce the number of false positives but at the cost of increasing the number of false negatives (missing true anomalies).[53] This trade-off between precision (the fraction of positive predictions that are correct) and recall (the fraction of actual positives that are found) must be carefully tuned based on the scientific goals and available resources.[53]
- **Cost-Sensitive Learning:** In applications where the cost of a false positive is different from the cost of a false negative, cost-sensitive learning can be employed. This involves assigning different penalties to different types of errors during the model training process, biasing the model to avoid the more "expensive" error.[53]
- **Rigorous Vetting Pipeline:** A multi-tiered vetting process, similar to that used in professional SETI searches, is essential.[31] A candidate "pebble" should first be checked against databases of known astrophysical phenomena and instrumental artifacts. It should then be subjected to re-observation to test for persistence. Only candidates that survive this rigorous filtration process should be elevated for deeper analysis.

The look-elsewhere effect and the false positive problem are, in essence, two facets of the same fundamental challenge. The former describes the statistical certainty of finding spurious signals, while the latter describes the act of misidentifying those signals as genuine. Together, they redefine the core task of the heuristic. The primary challenge is not simply anomaly *detection*, but anomaly *characterization and filtration*. The system must be designed not just to ask "Is this data point unusual?" but to rigorously answer the follow-up question: "What is the probability that this unusual data point was generated by a known natural process or instrumental effect, versus an unknown process of technological origin?" This reframes the entire endeavor from a simple search to a systematic process of evidence-based exclusion.

5.3 The Human-in-the-Loop: The Critical Role of Active Learning

The most sophisticated and efficient solution for navigating the false positive epidemic is to integrate human expertise directly into the machine learning loop through a process called "active learning".[45] This approach transforms the relationship between the human and the machine from a simple user-tool dynamic into a powerful collaborative partnership.

The active learning process works as follows:

1. The unsupervised machine learning model analyzes the bulk of the data and identifies a small subset of the most anomalous or, crucially, the most *ambiguous* candidates—those it is least certain how to classify.
2. This small, curated subset is presented to a human domain expert (e.g., an astronomer).
3. The expert provides labels for these few examples, classifying them as "interesting anomaly," "known instrumental artifact," "rare but known astrophysical object," etc.
4. This newly labeled data is fed back into the model, which retrains and updates its parameters.

This iterative feedback loop allows the model to learn from the expert's intuition and domain knowledge with extraordinary efficiency. Instead of requiring millions of labeled examples, the model's performance can be dramatically improved with just a few hundred intelligently chosen labels.[45] The

astronomaly framework, a real-world application of this concept to astronomical data, has demonstrated that incorporating active learning can nearly double the detection rate of scientifically interesting anomalies compared to using an unsupervised algorithm like Isolation Forest alone.[45]

This human-in-the-loop system creates a powerful positive feedback loop that extends beyond the search for technosignatures. Many of the anomalies flagged by the system will be false positives from a SETI perspective, but they are still, by definition, *real, rare, and unusual astrophysical phenomena*. As human experts label these objects—for instance, as a "rare type of quasar" or an "unusual stellar flare"—the machine learning model not only becomes more adept at filtering out non-technosignatures, but it also simultaneously builds a comprehensive, curated, and labeled dataset of rare natural events. This byproduct dataset is of immense value to the broader field of mainstream astronomy. A well-designed Alien Pebble Heuristic, therefore, functions as a dual-use discovery engine. Its primary objective is the speculative search for ETI, but its secondary, guaranteed outcome is the systematic discovery and classification of new and rare natural cosmic phenomena. This provides a robust scientific justification for the research program, ensuring a valuable return on investment

regardless of the outcome of the SETI search.

Section 6: Synthesis and Recommendations: A Practical Implementation Strategy

6.1 An End-to-End Architectural Proposal for the Alien Pebble Heuristic

To move from concept to practice, the principles discussed in this report can be synthesized into a concrete, end-to-end system architecture. This proposed architecture can leverage the design philosophies of existing open-source anomaly detection frameworks such as Orion, which provides a user-friendly library for time-series anomaly detection, and astronomaly, which pioneers the use of active learning in astronomical data exploration.[45] A modular, scalable system is envisioned, comprising the following core components:

1. **Data Ingestion and Preprocessing Module:** This module would be responsible for interfacing with major astronomical archives and real-time data streams. It would handle the normalization, cleaning (e.g., removing instrumental artifacts, cosmic rays), and feature engineering required to prepare diverse data types (images, spectra, light curves, catalogs) for analysis.[42]
2. **Multi-Stage Algorithmic Funnel (The Heuristic Engine):** This is the core analysis component, implementing the three-stage funnel described in Section 4. It would be a flexible pipeline allowing for the interchangeable use of various unsupervised algorithms (e.g., iForest, VAEs, LSTMs) tailored to specific data types and scientific questions.
3. **Candidate Vetting and Scoring Module:** Any anomaly flagged by the Heuristic Engine would be passed to this module. Here, it would undergo an automated vetting process. This includes a cross-check against catalogs of known variable stars, asteroids, and other potential contaminants. A statistical significance score would be calculated, explicitly correcting for the look-elsewhere effect based on the scope of the search that produced the candidate.
4. **Active Learning Interface:** High-scoring, persistent candidates that survive the vetting module would be passed to this human-in-the-loop interface. This user-facing tool would present the candidate data (e.g., image cutouts, light curves, spectra) to domain experts for labeling. The interface must be designed for efficiency, allowing experts to rapidly classify candidates and provide feedback to the Heuristic Engine for iterative

retraining.

5. **Data Fusion and Correlation Engine:** A crucial, advanced component that would actively search for correlated anomalies across different datasets. For example, if the Heuristic Engine flags a transient optical event in LSST data, this module would automatically query contemporaneous data from radio surveys (e.g., FAST) and spectral archives to search for corresponding anomalous behavior in the same sky location. A candidate that is anomalous across multiple, physically distinct domains would receive the highest priority score.

This architecture requires a significant cultural and infrastructural shift within the astronomical community toward open data standards and collaborative, cross-survey analysis platforms. The success of the heuristic is contingent not just on better algorithms, but on building a "data commons" where these algorithms can operate seamlessly across the world's premier astronomical datasets.

6.2 Prioritizing Target Datasets: Where to Search First

Given the vastness of the available data, a strategic prioritization of search targets is essential for maximizing the probability of a detection in the near term. Based on the "pebble" taxonomy and current instrumental capabilities, the following datasets represent the highest-priority targets:

- **Priority 1: Exoplanet Atmospheric Spectra.** The data currently being generated by the James Webb Space Telescope (JWST), and anticipated from future missions like the Habitable Worlds Observatory, represent the most promising avenue for detecting atmospheric "biomarkers" of technology.[33] A search for industrial pollutants like CFCs or other complex artificial molecules in the high-resolution spectra of terrestrial exoplanets is a direct, targeted test of the heuristic.[8]
- **Priority 2: High-Cadence, All-Sky Optical Surveys.** The forthcoming data from the Vera C. Rubin Observatory's LSST will provide an unprecedented dataset for detecting kinetic and transient anomalies.[33] Its deep, wide, and fast survey strategy is perfectly suited for identifying interstellar objects with non-gravitational acceleration, exoplanets with anomalous transit signatures (technofossils), or transient surface phenomena that could indicate technological activity.
- **Priority 3: Archival and Future Radio Data.** Decades of data exist from radio surveys like SERENDIP and Breakthrough Listen.[1] Re-analyzing these vast archives with the modern unsupervised algorithms proposed in this report could uncover subtle, structured, or wide-band signals that were missed by previous analyses focused on narrow-band beacons. Machine learning techniques are already being applied to this challenge with promising initial results.[55] The advent of the SKA will increase the volume

and sensitivity of this data domain by orders of magnitude.

The most valuable "pebbles" will likely be those that are anomalous across multiple of these domains simultaneously. A golden candidate would be an exoplanet that exhibits an anomalous atmospheric spectrum (a spectral pebble from JWST), a slightly non-Keplerian transit timing variation (a kinetic pebble from LSST), and a low-level, structured radio emission (an electromagnetic pebble from SKA). The proposed architecture must therefore be designed to prioritize the discovery of these high-confidence, multi-modal correlated anomalies.

6.3 Future Frontiers: The Potential of AI-Driven Hypothesis Generation and Quantum Sensing

Looking beyond the immediate implementation, the framework for the Alien Pebble Heuristic can be extended to incorporate next-generation technologies that promise to revolutionize the search.

- **AI-Driven Hypothesis Generation:** Current machine learning models are primarily used for pattern recognition and classification. However, the next frontier is in AI-driven scientific discovery. Future systems, potentially leveraging the architecture of Large Language Models (LLMs) or other forms of generative AI, could be tasked not just with finding anomalies but with analyzing them in the context of all known physics and proposing plausible hypotheses—both natural and artificial—to explain them.[33] Such a system could dramatically accelerate the interpretation phase of a discovery.
- **Quantum Sensing and Communication:** While highly speculative, the principles of quantum mechanics may offer entirely new domains for technosignatures. Theoretical work suggests that quantum coherence could be maintained over interstellar distances, potentially allowing for quantum communication.[4] The detection of such signals, or the use of advanced quantum sensors to probe for subtle physical effects not accessible to classical instruments, represents a future class of "pebbles" that the heuristic could one day be adapted to search for.[33]

Section 7: Conclusion: From Pebbles to Patterns, From Anomaly to Intelligence

7.1 Recapitulation of the Heuristic's Potential to Reshape SETI

The Alien Pebble Heuristic, as detailed in this report, represents a fundamental and necessary evolution in the Search for Extraterrestrial Intelligence. It advocates for a paradigm shift from a speculative and hopeful search for active, deliberate communication to a systematic, data-driven search for the passive and unintentional artifacts of technology. By focusing on subtle, contextually defined anomalies in the torrent of modern astronomical data, the heuristic opens up a vast and largely unexplored search space.

The proposed framework, grounded in the principles of unsupervised machine learning and inspired by proven methodologies in cybersecurity and diagnostic medicine, provides a practical pathway for this search. Its multi-stage algorithmic funnel is designed to efficiently process petabyte-scale datasets, while its integrated statistical vetting and human-in-the-loop active learning components provide the necessary rigor to navigate the perils of false positives and the look-elsewhere effect. This approach transforms SETI from a quest to find a single, extraordinary signal into a methodical process of elimination, much like any other branch of observational astronomy. It seeks to identify phenomena that cannot be explained by known natural processes, thereby isolating candidate technosignatures through a rigorous application of the scientific method. The dual-use nature of the proposed active learning system, which guarantees the discovery of rare natural phenomena as a byproduct of the search for artificial ones, provides a compelling justification for investment in this research program, promising tangible scientific returns irrespective of the ultimate success of the SETI component.

7.2 The Philosophical Implications of Searching for Traces Instead of Voices

The adoption of the Alien Pebble Heuristic carries with it profound philosophical implications that extend beyond mere search strategy. To detect a "pebble"—an artifact, a pollutant, a ruin—is fundamentally different from receiving an intentional "hello." It is an act of archaeology, not of communication.[2] This shift recasts our role from passive listeners waiting for a message to active explorers uncovering ancient or distant histories.

The consequences of such a discovery would be immense and complex. We might find evidence of a civilization that is currently thriving but entirely indifferent to or unaware of our existence. Alternatively, we could uncover the technofossils of a civilization that vanished

millions of years ago, providing a sobering data point for the longevity term in the Drake Equation and a stark warning about the potential fate of our own technological society.[21] The detection of a non-signal technosignature would not initiate a dialogue but would instead present humanity with a cosmic artifact to be studied, a puzzle to be solved. It would confirm that we are not alone, but it might simultaneously reinforce our profound isolation.

The questions raised by such a discovery—how to interpret the finding, whether to attempt a response, how to manage the societal impact—are complex and demand careful consideration long before a detection is made.[57] Ultimately, the Alien Pebble Heuristic is more than a technical tool; it is a new philosophical lens through which to conduct our search for life beyond Earth. It encourages a posture of patient, meticulous, and agnostic observation, reminding us that the first evidence of another intelligence may not be a shout from the stars, but a quiet whisper found within the data—a single, anomalous pebble on an infinite cosmic shore.

Works cited

1. Radio Searches - Extraterrestrial intelligence - Britannica, accessed on September 1, 2025, https://www.britannica.com/science/extraterrestrial-intelligence/Radio-searches
2. Archeology, Anthropology, and Interstellar Communication, accessed on September 1, 2025, https://www.chicagospace.org/archeology-anthropology-and-interstellar-communication/
3. Berkeley SETI, accessed on September 1, 2025, https://seti.berkeley.edu/listen/research.html
4. Search for extraterrestrial intelligence - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Search_for_extraterrestrial_intelligence
5. The Fermi Paradox revisited: Technosignatures and the Contact Era Amri Wandel, Racah Institute of Physics, The Hebrew Universit - arXiv, accessed on September 1, 2025, https://arxiv.org/pdf/2211.16505
6. (PDF) The Fermi Paradox revisited: Technosignatures and the Contact Era - ResearchGate, accessed on September 1, 2025, https://www.researchgate.net/publication/365889192_The_Fermi_Paradox_revisited_Technosignatures_and_the_Contact_Era
7. Fermi paradox - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Fermi_paradox
8. Technosignature - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Technosignature
9. The Search For Extraterrestrial Technosignatures: Expanding The Scope Of SETI With Advanced Computational And Quantum Sensors., accessed on September 1, 2025, https://quantumzeitgeist.com/the-search-for-extraterrestrial-technosignatures-expanding-the-scope-of-seti-with-advanced-computational-and-quantum-senso

rs/
10. Searching for Signs of Intelligent Life: Technosignatures - NASA Science, accessed on September 1, 2025, https://science.nasa.gov/universe/search-for-life/searching-for-signs-of-intelligent-life-technosignatures/
11. To find alien life, try eavesdropping on their conversations with spacecraft, astronomers suggest - Astronomy Magazine, accessed on September 1, 2025, https://www.astronomy.com/science/to-find-alien-life-try-eavesdropping-on-their-conversations-with-spacecraft-astronomers-suggest/
12. Our Radar Systems Have Accidentally Turned Earth into a Giant Space Beacon for the Last 75 Years and Scientists Say Aliens Could Be Listening - ZME Science, accessed on September 1, 2025, https://www.zmescience.com/science/physics/radar-systems-aliens-listening/
13. Civilian and Military Radar Leakage is Revealing Our Existence to Alien Civilizations, Study Suggests | Sci.News, accessed on September 1, 2025, https://www.sci.news/astronomy/civilian-military-radar-leakage-alien-civilizations-14055.html
14. Technosignatures and Astrobiology | Oxford Research Encyclopedia of Planetary Science, accessed on September 1, 2025, https://oxfordre.com/planetaryscience/display/10.1093/acrefore/9780190647926.001.0001/acrefore-9780190647926-e-275?p=emailAoAUpHc0Ykv56&d=/10.1093/acrefore/9780190647926.001.0001/acrefore-9780190647926-e-275
15. “Aliens Will Use It Against Us”: This Everyday Infrastructure Could Be the Real Reason Humanity Gets Contacted—or Conquered - Sustainability Times, accessed on September 1, 2025, https://www.sustainability-times.com/research/aliens-will-use-it-against-us-this-everyday-infrastructure-could-be-the-real-reason-humanity-gets-contacted-or-conquered/
16. What are the Detection Methods of IDS? - Stamus Networks, accessed on September 1, 2025, https://www.stamus-networks.com/blog/what-are-the-detection-methods-of-ids
17. Network Intrusion Detection System - Stamus Networks, accessed on September 1, 2025, https://www.stamus-networks.com/network-intrusion-detection-system
18. What Is Heuristic Analysis? Detection and Removal Methods | Fortinet, accessed on September 1, 2025, https://www.fortinet.com/resources/cyberglossary/heuristic-analysis
19. What Is Heuristic Analysis? - ITU Online IT Training, accessed on September 1, 2025, https://www.ituonline.com/tech-definitions/what-is-heuristic-analysis/
20. What is Heuristic Analysis? - Kaspersky, accessed on September 1, 2025, https://usa.kaspersky.com/resource-center/definitions/heuristic-analysis
21. Searching for Technosignatures: Implications of Detection and Non-Detection | Request PDF - ResearchGate, accessed on September 1, 2025, https://www.researchgate.net/publication/331840443_Searching_for_Technosignatures_Implications_of_Detection_and_Non-Detection

22. Biosignatures and Technosignatures – Astrobiology - CUNY Pressbooks Network, accessed on September 1, 2025, https://pressbooks.cuny.edu/astrobiology/chapter/technosignatures/
23. en.wikipedia.org, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Out-of-place_artifact#:~:text=An%20out%2Dof%2Dplace%20artifact,its%20presence%20in%20that%20context.
24. Out-of-place artifact - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Out-of-place_artifact
25. What Technosignatures Would Interstellar Objects Have? - Universe Today, accessed on September 1, 2025, https://www.universetoday.com/articles/what-technosignatures-would-interstellar-objects-have
26. Archaeology, Anthropology, and Interstellar Communication - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Archaeology,_Anthropology,_and_Interstellar_Communication
27. The Search for Extraterrestrial Artifacts: Evaluating the Possibility of Lost Civilizations Beyond Earth | by Boris (Bruce) Kriger | THE COMMON SENSE WORLD | Medium, accessed on September 1, 2025, https://medium.com/common-sense-world/the-search-for-extraterrestrial-artifacts-evaluating-the-possibility-of-lost-civilizations-beyond-d764b30e3381
28. Top Medical Advancements & Treatments - Narayana Health Blogs, accessed on September 1, 2025, https://www.narayanahealth.org/blog/understanding-biomarkers-a-beginner-s-guide-to-their-role-in-disease-detection
29. Biomarkers in clinical medicine - IARC Publications, accessed on September 1, 2025, https://publications.iarc.who.int/_publications/media/download/1412/33f111bfadb61211e9775dbf48c843c92f4220f9.pdf
30. Biomarkers: What Are They? - WebMD, accessed on September 1, 2025, https://www.webmd.com/a-to-z-guides/biomarkers-overview
31. A 2821 Star Optical SETI Survey using ESO HARPS archival data - arXiv, accessed on September 1, 2025, https://arxiv.org/html/2508.08628v1
32. 10 Uses Of Data Science In Space Exploration | by Emerging India Analytics | Medium, accessed on September 1, 2025, https://medium.com/@analyticsemergingindia/10-uses-of-data-science-in-space-exploration-d958045213c2
33. SETI Post-Detection Futures: Directions for Technosignature Research and Readiness, accessed on September 1, 2025, https://arxiv.org/html/2507.11587v1
34. Exoplanetary Atmospheres—Chemistry, Formation Conditions, and ..., accessed on September 1, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC5207327/
35. Planetary And Exoplanetary Atmospheres - JPL Science - NASA, accessed on September 1, 2025, https://science.jpl.nasa.gov/division/planetary-science/planetary-and-exoplanetary-atmospheres/

36. Signal-to-noise ratio - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Signal-to-noise_ratio
37. Calculating Signal-to-Noise Ratio Using DFT - NumXL, accessed on September 1, 2025, https://numxl.com/blogs/calculating-signal-to-noise-ratio-using-dft/
38. A Systematic Review of Anomaly Detection within High ... - JOIV, accessed on September 1, 2025, https://joiv.org/index.php/joiv/article/viewFile/1297/641
39. Anomaly Detection Framework for Wearables Data: A Perspective Review on Data Concepts, Data Analysis Algorithms and Prospects - MDPI, accessed on September 1, 2025, https://www.mdpi.com/1424-8220/22/3/756
40. Identifying Outliers in Astronomical Images with Unsupervised Machine Learning - Research in Astronomy and Astrophysics (RAA), accessed on September 1, 2025, http://www.raa-journal.org/issues/all/2022/v22n8/202208/P020220805283381824834.pdf
41. Anomaly Detection in Large-Scale Log Data - DiVA Portal, accessed on September 1, 2025, https://umu.diva-portal.org/smash/get/diva2:1983804/FULLTEXT01.pdf
42. Anomaly Detection in Astronomical Data - Imarticus Learning, accessed on September 1, 2025, https://imarticus.org/blog/anomaly-detection-in-astronomical-data/
43. Review of Anomaly Detection Algorithms for Data Streams - MDPI, accessed on September 1, 2025, https://www.mdpi.com/2076-3417/13/10/6353
44. anomaly-detection-framework - PyPI, accessed on September 1, 2025, https://pypi.org/project/anomaly-detection-framework/
45. Astronomaly at scale: searching for anomalies amongst 4 million galaxies | Monthly Notices of the Royal Astronomical Society | Oxford Academic, accessed on September 1, 2025, https://academic.oup.com/mnras/article/529/1/732/7612998
46. Anomaly Detection Methods for Finding Technosignatures - SciTePress, accessed on September 1, 2025, https://www.scitepress.org/Papers/2024/123064/123064.pdf
47. Anomaly Detection in Astrophysics: A Comparison Between Unsupervised Deep and Machine Learning on KiDS Data - IRIS Unina, accessed on September 1, 2025, https://www.iris.unina.it/retrieve/b215b889-22aa-4d43-88fc-4faba258dfdd/163-DAddona-978-3-030-65867-0_10.pdf
48. An anomaly detection framework anyone can use | MIT News, accessed on September 1, 2025, https://news.mit.edu/2025/anomaly-detection-framework-anyone-can-use-sarah-alnegheimish-0528
49. Self-calibrating the look-elsewhere effect: fast evaluation of the statistical significance using peak heights | Monthly Notices of the Royal Astronomical Society | Oxford Academic, accessed on September 1, 2025, https://academic.oup.com/mnras/article/508/1/1346/6373472
50. Look-elsewhere effect - Wikipedia, accessed on September 1, 2025, https://en.wikipedia.org/wiki/Look-elsewhere_effect
51. [2007.13821] The look-elsewhere effect from a unified Bayesian and frequentist

perspective, accessed on September 1, 2025, https://arxiv.org/abs/2007.13821
52. Glossary: Data Science - False Positive Rate - C3 AI, accessed on September 1, 2025, https://c3.ai/glossary/data-science/false-positive-rate/
53. False Positives and False Negatives - GeeksforGeeks, accessed on September 1, 2025, https://www.geeksforgeeks.org/machine-learning/false-positives-and-false-negatives/
54. Artificial Intelligence, Exoplanets, & the Future of Astrobiology | Ask an Astrobiologist, accessed on September 1, 2025, https://www.youtube.com/watch?v=Lt50527eHjU
55. SETI's new machine learning algorithm works like Google's "reverse image search" on a galactic scale, and has already uncovered eight currently unexplained signals. Researchers train the algorithm on simulated alien signals, then look back through old data to see if we missed anything in the noise. : r/space - Reddit, accessed on September 1, 2025, https://www.reddit.com/r/space/comments/123oz7i/setis_new_machine_learning_algorithm_works_like/
56. Breakthrough Listen: Deep Learning Search for Technosignatures - Berkeley SETI, accessed on September 1, 2025, https://seti.berkeley.edu/ml_gbt/
57. Seeking Signs: If Aliens Explore Space As We Do, We Should Listen for Their Calls to Other Planets - BIOENGINEER.ORG, accessed on September 1, 2025, https://bioengineer.org/seeking-signs-if-aliens-explore-space-as-we-do-we-should-listen-for-their-calls-to-other-planets/
58. How Should NASA Respond If We Found An Alien Technosignature? - IFLScience, accessed on September 1, 2025, https://www.iflscience.com/how-should-nasa-respond-if-we-found-an-alien-technosignature-80106
59. [1903.06550] Searching for Technosignatures: Implications of Detection and Non-Detection - arXiv, accessed on September 1, 2025, https://arxiv.org/abs/1903.06550